Iodine Doping Facilitates the Phase Formation of Lithium-Argyrodite Solid Electrolyte

Abstract: All-solid-state lithium batteries (ASSLBs) with solid-state electrolytes (SSEs) have attracted wide attention for their high energy density and reliable safety performance. Halogen-substituted lithium argyrodites, Li_6PS_5Cl (LPSC), represent class of promising and suitable sulfide-based solid electrolytes with high conductivity and great processability. In the mass production process of simple high-speed mixing of raw materials without treatment, however, obtained LPSC electrolyte will still exist Li_2S raw material signal. Herein, in this study, introducing LiI to argyrodite LPSC electrolyte is proposed to reduce the raw material phase of Li_2S and enhance ionic conductivity (up to 3.91 $mS \cdot cm^{-1}$). Moreover, iodization of the argyrodite LPSC is found lower the crystallization temperature of LPSC. By combining $Li_6PS_5Cl_{0.95}I_{0.05}$ (I0.05-LPSC) with a graphite (Gr) anode and a NCM622 cathode, a room temperature ASSLBs with excellent long-term cycling stability (97.9% capacity retention at 0.3 C for 100 cycles) and a high reversible capability (134.8 $mAh \cdot g^{-1}$ at 0.3 C) is reported. These results prove that the doping of LiI can reduce the difficulty of synthesis of argyrodite.

Keywords: lithium argyrodites, Li_6PS_5Cl, Li_2S raw peak, LiI, ionic conductivity, solid state battery

1. Introduction

Lithium-ion batteries (LIBs) have occupied the consumer electronics market, dominating the lead in electric vehicles, and energy storage[1-7]. Due to the flammable property of organic electrolytes, commercial lithium-ion batteries suffer a serious thermal runaway problem, which prevents its further application and development[8-10]. All-solid-state lithium batteries (ASSLBs) with non-flammable inorganic solid electrolytes are promising alternatives to the safety problem. Solid state electrolytes (SSEs) have significantly drawn attention due to their wide electrochemical voltage window, excellent thermal stability, good manufacturing performances, wide operating temperature range, and superior safety[11-13].

Specifically, sulfide solid electrolytes are strong candidates for all-solid-state batteries due to their high ionic conductivity compared to other electrolytes[14, 15]. The argyrodites Li_6PS_5X (X = Cl, Br) with high ionic conductivity of$\sim 10^{-3}$ S cm^{-1} and low cost owing to the utilization of cheap synthesis precursors are widely studied in recently years, which is a representative in sulfide solid electrolytes[16, 17]. Some previous studies have demonstrated that their conductivities are affected by lithium vacancy concentration and structural disorder between the S^{2-}/X^{-} positions[18-20]. Among them, several reports have been published on the enhancement of conductivity by cationic and anion doping[16, 21, 22].

In a general way, the most widely used method of preparing sulfide SSEs is high-energy ball milling followed by annealing. The high-energy ball milling process is time-consuming and the yield is low[23-26], which is not reliable for mass production. If the direct mixing method is used, large particle size of precursor will lead to a residual Li_2S signal[27]. These residua Li_2S can cause the conductivity of the electrolyte to still not be high enough. In previous research, Ujiie Satoshi et al. [28] reported that addition of LiI decreased the intensity of the X-ray diffraction (XRD) peak attributable to 70Li2S·30P2S5 glass–ceramic and improved the ionic conductivity.

In this work, completely untreated raw materials are used, and all the raw materials are simply mixed and then sintered. After doping LiI, the sintering temperature is reduced (550 to 480°C), and the Li_2S raw material peak in electrolyte is less. The $Li_6PS_5Cl_{0.09}I_{0.05}$ electrolytes present the high room temperature ion conductivity of 3.91 $mS \cdot cm^{-1}$. In-situ X-ray diffraction (in-situ XRD) was used to monitor the sintering process, it was found that the phase forming temperature of argyrodite decreased. Finally, the NCM622/ $Li_6PS_5Cl_{0.95}I_{0.05}$ /graphite ASSLBs was used to study the electrochemical properties of $Li_6PS_5Cl_{0.95}I_{0.05}$ electrolyte.

2. Experimental Section

2.1 Preparation of solid electrolytes

To prepare $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolytes, Li_2S (99.98%, GanFeng Lithium Co., Ltd.), P_2S_5(99%, P > 24 wt%, Sinopharm Chemical Reagent Co., Ltd.), and anhydrous LiCl (99.98%, dried at150 °C, <200 ppm, Aladdin) crystalline powders were weighed according to a certain stoichiometric ratio of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4). Then the 20g raw materials were pulverized at ultra-high speed (25000 rpm) with a powder-mixing machine (DaDeYaoJi)[27]. The precursor powder sample with uniform particle sizes can be obtained in 150 s by a total of 6 stirring times (25 s per time), following 1 minute interval each time for heat dissipation of the powder-mixing machine. The obtained precursor powder was sealed in a quartz bottle and annealed at 480 °C (heating rate of 2 °C min^{-1}) for 10 h and then naturally cooled to obtain the final $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolytes. All the above operations were carried out under an argon atmosphere.

2.2 Materials characterization

Powder XRD patterns were collected using an Ultima IV. XRD cells containing hermetically sealed SSE samples with a polyimide (PI) film were mounted on an XRD diffractometer and measured at 40 kV and 40 mA from 10 ° to 70 ° with a scan speed of 2.4 ° min^{-1}. The powder was tested by XRD in situ, and the test temperature was 25-

550 °C, and the sample was taken every 30 °C, and the holding time at each temperature point was 30 min. Rietveld refinements of Powder XRD data were conducted using Fullprof software. X-ray photoelectron spectroscopy (XPS) was performed by PHI 5000 Versa Probe III with a monochromatic Al Kα X-ray source with a beam size of 200 um using inert-atmosphere protected transferring device. The observed binding energies were calibrated with the adventitious C_{1s} peak to 284.8 eV. Scanning electron microscopy (SEM, TESCAN MIRA3) equipped with an energy dispersive spectroscopy (EDS, Bruker) addon was adopted to characterize the morphologies and elemental distributions, respectively. Raman spectra were measured with AW200SG, Electrotek Raman spectrophotometer (UK) in the range of 50-450 cm^{-1}, using an Ar ion laser beam and an excitation wave length of 514 nm.

2.3 Electrochemical characterization

The ionic conductivity of the $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolyte material was analyzed by AC impedance spectroscopy. The tested electrolyte pellets were obtained by pressing 300 mg of electrolyte powder into a pellet with a 10 mm diameter under a pressure of 1.25 tons. The carbon-coated aluminum foils were placed on both sides of the electrolyte pellet as the blocking electrode. AC impedance test was carried out using Ivium electrochemical workstation in the frequency range of 1 Hz to 1 MHz with an applied voltage of 0.2 V. The In/ SSEs /In symmetric batteries are used for EIS at different temperatures, which vary continuously from low to high temperature (-15~75 °C) by Toyo Co. Ltd. LN-Z2-HF.

All solid-state NCM622/I0.05-LPSC/Gr mold batteries were assembled in the following manner. Commercial NCM622 and I0.05-LPSC electrolyte were prepared in a weight ratio of 7:3 into a mixer for high-speed mixing (Thinky mixer ARE310) at a speed of 1000 rpm for 150 s to obtain the final composite cathode. The preparation process of the composite anode is similar to that of the cathode, except that the weight ratio of graphite and electrolyte is 6:4. During the assembly of the full battery, we first obtain electrolyte pellets by pressing 100 mg of electrolyte powder into a pellet with a

10 mm diameter under a pressure of 1.25 tons. Then 21 mg of the above-mentioned cathode was used in combination with 100 mg of I0.05-LPSC electrolyte to make a 10 mm two-layer pellet by a pressure of 1.9 tons. Next, a pressure of 2.5 tons was applied to make the three-layer pellet followed by attaching 14 mg anode mixture on the other side. Finally, we assembled the battery into a mold. All the electrochemical properties measurements for solid-state lithium batteries in this work are performed at room temperature.

3. Results and Discussion

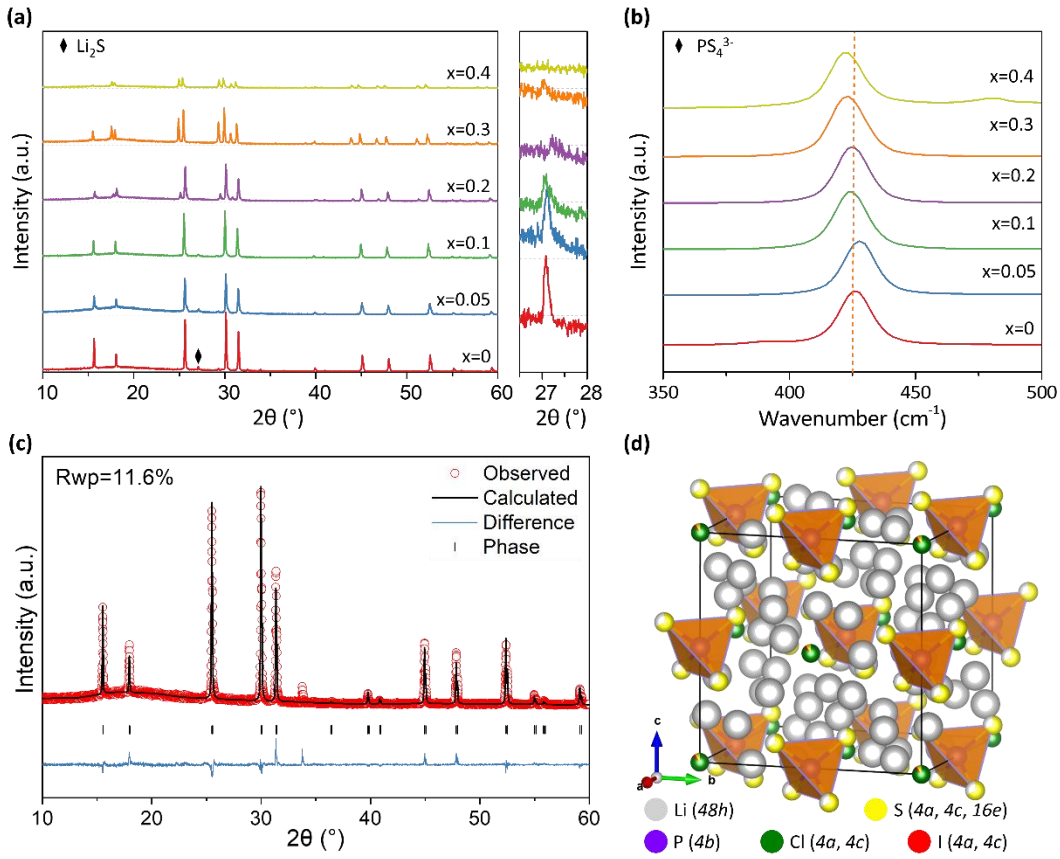

Figure 1. (a) Measured X-Ray diffraction patterns of the as-prepared $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) samples and the magnified angle on the right shows the main peak of lithium sulfide. (b) Raman spectra of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) samples. (c) XRD pattern of and the corresponding Rietveld refinement of $Li_6PS_5Cl_{0.95}I_{0.05}$ sample. (d) Structural representation of the $Li_6PS_5Cl_{0.95}I_{0.05}$ samples.

Figure 1a depicts XRD patterns of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolytes, the main peaks at 30° could be indexed as cubic argyrodite phase (F-43m)[29], indicate that the argyrodite electrolyte with fast ion channels can be synthesized in this way. However, there is a raw material peak of Li_2S in the LPSC phase, and after doping LiI, the peak intensity of Li_2S gradually decreases with the increase of the amount of incorporation. But if the doping is too much (x > 0.1), the main peak splitting will occur due to the phase change caused by the lattice constant. In addition, Figure 1b shows the Raman spectroscopy results of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolyte, indicating that the peak at 425 and 568 cm^{-1} corresponds to the PS_4^{3-} group, which further testify the argyrodite structure[30]. Besides, with the I concentration increased, the main PS_4^{3-} unit appear no split and slightly shifts the wave number towards the lower direction (red shift)[31, 32]. These results indicate that I entered the lattice of S-Cl of the argyrodite structure.

In order to gain a deeper understanding of the impact of doping, XRD Rietveld refined of $Li_6PS_5Cl_{0.95}I_{0.05}$ sample was used. The fitted pattern and refined pattern parameters are shown in Figure 1c and Table S1, respectively. On account of the large radius of I than that of Cl, the appearance of the lattice expansion phenomenon is detected by the refined larger lattice parameter of a=9.874Å, with comparison to a=9.858Å for Li_6PS_5Cl structure (Figure S1 and Table S2). The Rietveld analysis also suggests that the incorporated I atoms share the 4a/4c site with Cl atoms and present an occupation ratio of 4.99%, which is very close to 5% doping amount of previous design. Figure 1d shows the crystal structure of $Li_6PS_5Cl_{0.95}I_{0.05}$, in which the Cl^-/I^- anions form a face centered cubic lattice (Wyckoff positions 4a and 4c), with S^{2-} anions in half of the tetrahedral voids and PS_4^{3-} tetrahedra on the octahedral sites (P on Wyckoff 4b).

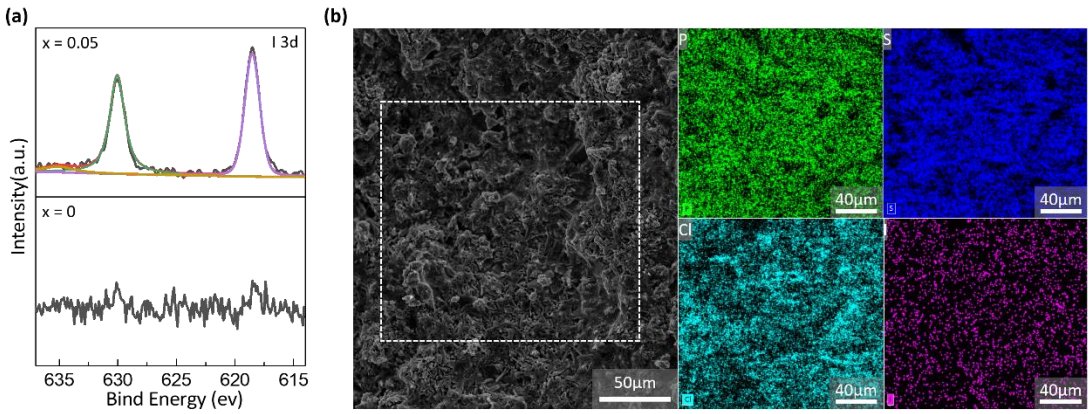

Figure 2. (a) Measured I 3d XPS spectra of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05) samples. (b) EDS mapping of $Li_6PS_5Cl_{0.95}I_{0.05}$ electrolyte pellet.

The prepared solid electrolyte was examined using XPS spectra. Figure S2a depicts the deconvolution spectra of the Li1s region of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolytes, and one sharp peak is clearly visible at 55.68 eV, which is correlated to Li1s from argyrodites. Also, the P 2p and S 2p spectra of the electrolytes show the characteristic signals for PS_4^{3-} tetrahedra[33]. Additionally, the XPS spectrum of the precursor LiI for I 3d is at 618.90 eV[34], and $Li_6PS_5Cl_{0.95}I_{0.05}$ electrolyte for I 3d appears at 618.18 eV, which is consist with the XPS spectrum of Li_6PS_5I[35]. These results indicate that the chemical state of I 3d has changed and the substituted I element existed in the LPSC structure. EDS mapping images of I0.05-LPSC and LPSC electrolyte pellet in Figure 2b and Figure S3b, I element was evidently detected in the EDS mapping of I0.05-LPSC, where P, S, Cl, and I were uniformly distributed. Furthermore, I0.05-LPSC electrolyte powder shows in Figure S3a, in which most of electrolyte particles display small and uniform particle size (<10 μm).

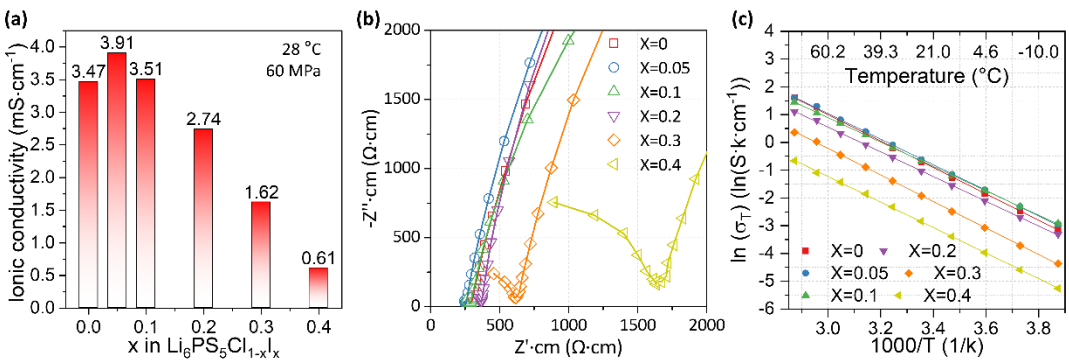

Figure 3. (a) Ionic conductivities measured at 28 °C. (b) Nyquist plots and the fitting plots, and (c) Arrhenius plots of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4).

The LPSC electrolyte is synthesized in Figure S4, and the sintering temperature needs to 550°C to achieve ionic conductivity of 4.3 $mS \cdot cm^{-1}$. Because the melting point of LiI is lower (469 °C), the electrolyte prepared in this study can obtain a high conductivity when the sintering temperature is 480 °C. The Li^+ conductivities of electrolyte were measured using cold-pressed pellets. Figure 3a shows the ionic conductivities of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) electrolytes. Clearly, the highest room temperature conductivity of 3.91 $mS \cdot cm^{-1}$ was obtained by $Li_6PS_5Cl_{0.95}I_{0.05}$. Conductivities of $Li_6PS_5Cl_{1-x}I_x$ electrolytes with x=0, 0.1, 0.2, 0.3, 0.4 are 3.47, 3.51, 2.74, 1.62 and 0.61 $mS \cdot cm^{-1}$, respectively. The corresponding Nyquist plots are presented in Figure 3b and the fitting parameters for $Li_6PS_5Cl_{1-x}I_x$ impendence spectra are shown in Table S3. The activation energy (*Ea*) was calculated using the equation $\sigma_T=\sigma_0 exp(-Ea/k_BT)$, where σ represents the Li^+ conductivity, σ_0 denotes a pre-factor, *T* represents the absolute temperature, and k_B is the Boltzmann constant[36]. The Arrhenius plots of the $Li_6PS_5Cl_{1-x}I_x$ in the temperature range from -15 to 75 °C are depicted in Figure 3c, and the calculated activation energy values of $Li_6PS_5Cl_{1-x}I_x$ electrolytes with x = 0, 0.05, 0.1, 0.2, 0.3,0.4 are 0.3958, 0.3972, 0.3783, 0.3838, 0.4117 and 0.3959 eV, respectively. Figure S5 shows the EIS results of $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2, 0.3, 0.4) under different temperatures.

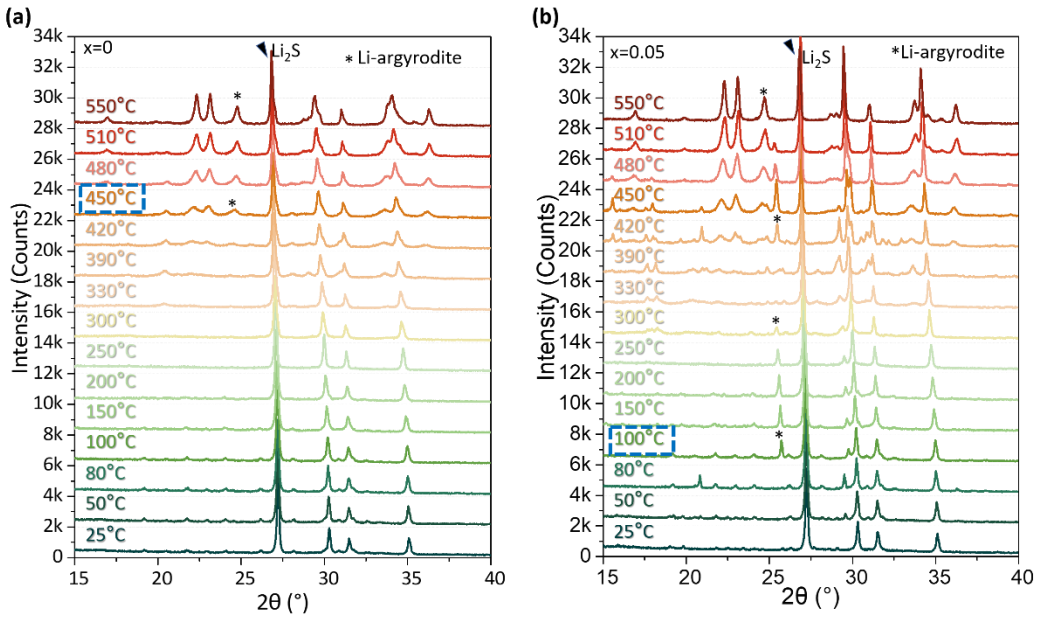

Figure 4. Details from *in-situ* XRD patterns at various temperatures for (a) Li_6PS_5Cl and (b) $Li_6PS_5Cl_{0.95}I_{0.05}$.

According to the results of the conductivity result, the doping limit was reached at $Li_6PS_5Cl_{0.8}I_{0.2}$. Therefore, $Li_6PS_5Cl_{1-x}I_x$ (x=0, 0.05, 0.1, 0.2) precursor mixture was subjected to in-situ XRD test in order to understand the effect of LiI doping on the preparation. Figure 4a shows in situ XRD data recorded during the heat treatment of Li_6PS_5Cl precursor. With the increase of temperature (25-550 °C), the characteristic peaks of Li-argyrodite phase appear at 450 °C and the peak intensity gradually increases, in addition, the peak intensity of Li_2S has maintained a high level. Similarly, the same test conditions were used for the $Li_6PS_5Cl_{0.95}I_{0.05}$ in Figure 4b. There are some different situations, the characteristic peak of Li-argyrodite phase began to appear at 100 °C, became very weak at 300 °C, but gradually increased after 420°C. Moreover, compared with Li_2S, the main characteristic peak intensity of Li-argyrodite is higher, and it takes up the main position in phase composition. A same situation also appears in $Li_6PS_5Cl_{0.9}I_{0.1}$ (Figure S6a) and $Li_6PS_5Cl_{0.8}I_{0.2}$ (Figure S6b), Li-argyrodite phase also first appear at 100°C, and more obviously, with the increase of LiI doping amount, and the peak intensity of Li_2S gradually decreases. The above results are similar to those obtained by predecessors[28, 37]. The above results indicate that doping LiI can help

to form the phase at a lower temperature and reduce the raw material phase of Li_2S in the argyrodite.

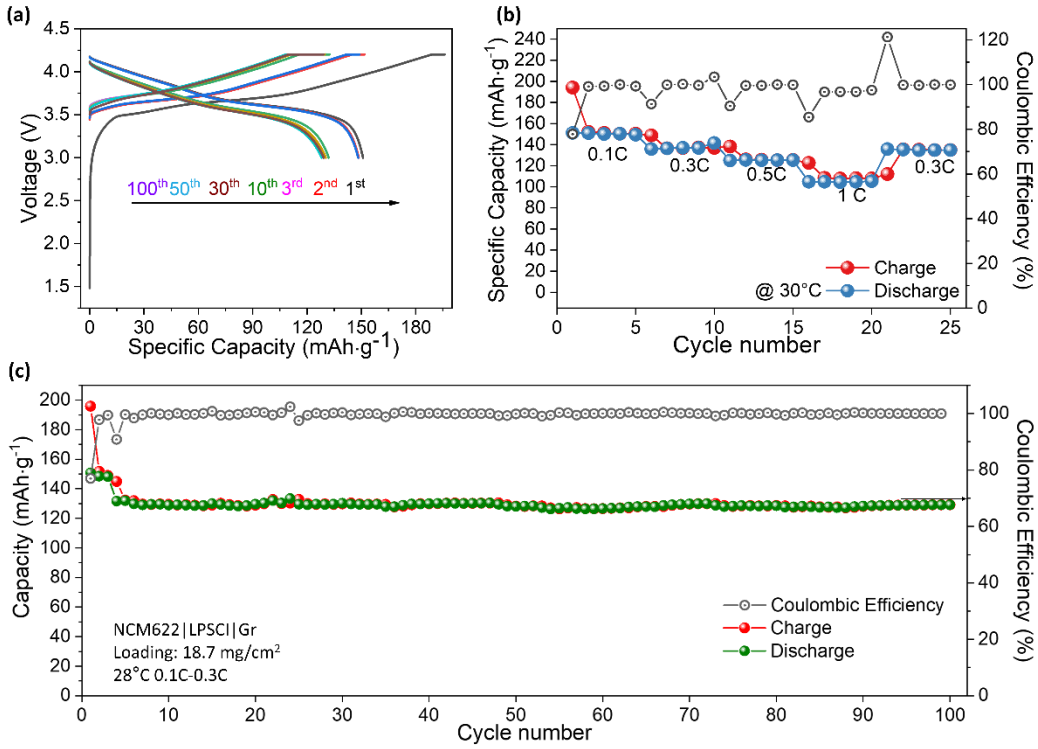

Figure 5. (a) The cycling performance, (b) charge-discharge curves, (c) rate performance of NCM622/ $Li_6PS_5Cl_{0.95}I_{0.05}$ /graphite ASSLBs.

In order to know the electrochemical properties of the I0.05-LPSC SSEs, the all-solid-state batteries by combining NCM622 cathode and graphite anode with $Li_6PS_5Cl_{0.95}I_{0.05}$ were assembled to assess the performance of the electrolytes. All of the batteries were tested by charging and discharging between 3 and 4.2 V at the loading of 18.7 mg cm^{-2}. The cycling performance of the NCM622/ I0.05-LPSC /Gr battery at various C-rates are presented in Figure 5a. The reversible discharge capacities for NCM622/ I0.05-LPSC /Gr battery at 0.1, 0.3, 0.5 and 1 C are 151.1, 135.8, 124.9, 105.2 $mAh \cdot g^{-1}$, respectively, then a capacity of 134.8 $mAh \cdot g^{-1}$ can be obtained at 0.3C, demonstrating excellent rate performance. Figure 5b displays the charge-discharge curves of NCM622/ I0.05-LPSC /Gr ASSLB, the charge capacity of the first circle is 195.8 $mAh \cdot g^{-1}$ at 0.1 C, the specific discharge capacity is 150.8 $mAh \cdot g^{-1}$, and the

Coulomb efficiency is 77%. Finally, figure 5c shows the long cycling stability of ASSLB, the charge and discharge test of the first three circles at 0.1 C (0.28 $mA \cdot cm^{-2}$), and the next charge and discharge test at 0.3 C, the battery exhibits a specific discharge capacity of 128.9 $mAh \cdot g^{-1}$ during a long cycle, and a capacity retention rate of 97.9% at 0.3C after 100 cycles.

4. Conclusions

In summary, in the case of direct powder mixing without raw material treatment, LPSC synthesis will produce a large Li_2S impurity peak, but the addition of LiI will effectively inhibit the production of impurity phases. Moreover, without I doping, LPSC needs to be heated to 550 ℃ to obtain high conductivity of 4.3 $mS \cdot cm^{-1}$, but after adding only 5% mol LiI, this temperature is reduced to 480 ℃, and the $Li_6PS_5Cl_{0.09}I_{0.05}$ electrolyte with a conductivity of 3.91 $mS \cdot cm^{-1}$ is obtained. In situ XRD test shows that the argyrodite LPSC phase of the LiI-doped precursor has already appeared at 100 ℃, while the original LPSC requires up to 450 ℃, indicating that LiI greatly reduces the phase forming temperature of LPSC synthesis. These results indicate that the doping of LiI can promote synthesis of the argyrodite LPSC. Furthermore, the obtained $Li_6PS_5Cl_{0.95}I_{0.05}$ electrolyte was employed in NCM622/ I0.05-LPSC /Gr batteries, delivering a 97.9% retention rate after 100 cycles. Therefore, these results prove that a small amount of LiI doping is a promising strategy in the preparation of large quantities to LPSC.

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